Transcendence of shifted Mills minima for odd multipliers up to one thousand

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Abstract

For a positive integer r, let ξ_r be the least real $A > 1$ for which every $\lfloor A^{r3^k-1} \rfloor$, $k \geq 1$, is prime. We prove that ξ_r is transcendental for every odd $r \leq 1000$. The case $r = 273$ was expressly left undecided by the proof in a previous corpus manuscript. Starting from Saito's Pisot reduction, a local trace identity confines an algebraic minimum at the exceptional multiples of thirteen to cubics of a specific form. Primality at an early, prescribed exponent bounds their coefficients. An exhaustive covering by residue classes then gives a recurring prime divisor for every surviving cubic. A standalone integer-arithmetic verification is included.

1 The question and the result

The starting index matters for a least prime-representing constant. Throughout, we keep every exponent in the definition

$$C_k = r3^k - 1 \quad (k \geq 1), \qquad \xi_r = \min\{A > 1 : \lfloor A^{C_k} \rfloor \text{ is prime for every } k \geq 1\}. \tag{1}$$

The minimum exists by Saito's Theorem 1.3, since $C_1 \geq 2$ and the successive ratios exceed three [1]. Saito proved transcendence unconditionally for $r \geq 4.003 \cdot 10^{14}$, and for the whole family under the Riemann hypothesis. His result identifies a specific number at each r, rather than merely producing some admissible transcendental number.

Theorem 1. *For every odd integer $1 \leq r \leq 1000$, the number ξ_r in (1) is transcendental. In particular, the least $A > 1$ for which $\lfloor A^{273 \cdot 3^k - 1} \rfloor$ is prime at every $k \geq 1$ is transcendental.*

Keio's corpus manuscript [2] derives transcendence for odd $13 \nmid r$ and also treats the two fixed shifts $3^k \pm 1$. Manuscript [3] handles the additional multipliers $13, 39, 65, 91, 195$. Its discussion after the 195 argument singles out 273. Although the 273 sequence is a tail of the 91 sequence, removing the first prime constraint can change the minimum. In particular $9 \cdot 91 - 1 = 818$, the second exponent for 91, is the first for 273, while $3 \cdot 91 - 1 = 272$ has disappeared. The proof below uses primality at an index actually present for 273. The theorem supplies a common verification for all 38 odd multiples of thirteen in the stated range, 33 beyond the five cases above.

The local return at three, the cubic descent, the parity restriction, and the height estimate come from [2, 3]. We prove them here from Saito's results and supply a uniform exhaustive final step. No implication from transcendence of either fixed minimum to transcendence of a different minimum is assumed. All references to Saito's numbering use arXiv version 3.

2 A cubic from the prime floors

Suppose r is odd and $A = \xi_r$ is algebraic. Put $B = A^2$ and $m_k = C_k/2$. Squaring gives an increasing bijection of the respective admissible sets, so B is the minimum for (m_k). We have

$m_{k+1} = 3m_k + 1$. For every i some $h > 0$ satisfies $3^h \equiv 1 \pmod{2m_i}$, since $2m_i = r3^i - 1$ is prime to three. Thus m_i properly divides m_{i+h}. An integral B^{m_i} would make the prime floor at $i + h$ a proper power.

Apply Saito's Proposition 3.1 with his unconditional Theorem 1.6 and $\theta = 21/40$. The exponent ratios exceed three and tend to three, so all sufficiently large indices belong to his index set. The power B^g in that proposition has $g \mid m_k$ eventually, hence $g = 1$. Its degree is at most $1 + 40/17 < 4$ and is at least two. Degree two would require all sufficiently late m_k to be odd primes, whereas m_k is even infinitely often. Consequently B is a cubic Pisot number. Writing its conjugates as $B_1 = B, B_2, B_3$, we have

$$P_k = \lfloor B^{m_k} \rfloor = \sum_{i=1}^{3} B_i^{m_k} \qquad (k \text{ sufficiently large}).$$

Here a Pisot number is a real algebraic integer greater than one whose other conjugates have modulus less than one. If B_2, B_3 were both real, their contributions at every even exponent would have positive sum, contrary to the floor identity. Thus $\{B_2, B_3\} = \{Re^{i\theta_0}, Re^{-i\theta_0}\}$ with $0 < \theta_0 < \pi$ and $0 < R < 1$.

We will need the quantitative bound $R \le B^{-17/40}$. For each $0 < \kappa < 17/40$ the fractional-part estimate in Saito's Proposition 3.1(i) gives

$$|2R^{m_k} \cos(m_k\theta_0)| = O_\kappa(B^{-\kappa m_k}), \qquad (1 - 21/40) \lim m_{k+1}/m_k - 1 = 17/40.$$

If $R > B^{-\kappa}$ then $\cos(m_k\theta_0) \to 0$. It follows from the triple-angle formula that $\cos(3m_k\theta_0) \to 0$ and $|\sin(3m_k\theta_0)| \to 1$. But $|\cos(m_{k+1}\theta_0)| = |\cos(3m_k\theta_0 + \theta_0)| \to \sin\theta_0 > 0$. This proves the bound by letting κ increase to $17/40$.

3 The local identity at three

Choose square roots $\gamma_i^2 = B_i$ in a fixed normal number field K. The eventual, increasing primes are $P_k = \sum_i \gamma_i^{r3^k - 1}$. Discard finitely many P_k that divide $3\prod_i N_{K/\mathbb{Q}}(\gamma_i)$. At a prime of K above P_k, let $d \le [K : \mathbb{Q}]$ be the residue degree and $E = P_k^d - 1$. If $v_3(E) \le k + v_3(r)$, there exists $h > 0$ with $r3^{k+jh} \equiv r3^k \pmod{E}$ for all $j \ge 0$, by taking 3^h congruent to one modulo the prime-to-three part of E. All γ_i are residue-field units, so $P_k \mid P_{k+jh}$, a contradiction for large j. Thus $v_3(P_k^{2d} - 1) > k + v_3(r)$. Since $v_3(P_k^{2d} - 1) = v_3(P_k^2 - 1) + v_3(d)$ and d is bounded, we get

$$\max\{v_3(P_k - 1), v_3(P_k + 1)\} \ge k - O_K(1).$$

Fix a place of K above three with residue degree f. For its set U of unit roots γ_i, let ω_i be the Teichmüller representative of the residue and put $v_i = \omega_i^r$. Along $k \equiv j \pmod{f}$ as $k \to \infty$, the powers of γ_i/ω_i with exponent $r3^k$ tend to one. Nonunit terms tend to zero. Since P_k approaches the set $\{-1, 1\}$, we obtain for every $j \ge 0$ an equality in algebraic numbers

$$E_j = \sum_{i \in U} v_i^{3^j}/\gamma_i \ \in \{-1, 1\}. \tag{2}$$

The representatives are roots of unity and hence algebraic. Thus the equalities in the completion are transported through any complex embedding of the resulting enlarged number field.

The set U cannot be empty. If it is a proper subset, take an automorphism of a common normal number field sending one of the B_i indexed by U to a nonunit root. Such an automorphism exists because the cubic is irreducible. When $|U| = 1$ there is just one transformed term. When $|U| = 2$ there is just one nonunit among the three roots, so exactly one transformed term has negative valuation. All numerators are roots of unity. In either case the sum has negative valuation, contrary to (2). Consequently all roots are units at three.

4 The cubic modulo three

Let $F = X^3 - uX^2 + vX - w$ be the polynomial of B. If F is reducible modulo three, all its residual roots belong to $\mathbb{F}_9^\times$. When they all belong to $\mathbb{F}_3^\times$, the numbers v_i^2 are signs. If these signs agree, the eventual traces $\sum B_i^{m_k}$ vanish modulo three, since their three residues coincide (r is odd). Otherwise one sign occurs once. One of $E_0 + E_1, E_0 - E_1$ isolates twice that term, since $v_i^3 = v_i v_i^2$. The isolated term is nonzero, so its doubled value is ± 2. This makes one γ_i a root of unity.

Otherwise there is one linear root and a conjugate quadratic pair of order either eight or four. If the quadratic roots have order eight, their v_i have order sixteen, while the linear v_i has order at most four. Hence $E_0 + E_2$ isolates the linear term, with the same contradiction since $v_i^9 = -v_i$ on the quadratic terms and $v_i^9 = v_i$ on the linear term. If the quadratic roots have order four then the two values v_i^2 there are negatives of each other. The determinant of their columns in the system with rows E_0, E_1 has modulus two in every complex embedding, since the determinant is $v_2 v_3 (v_3^2 - v_2^2)$ after indexing the pair as $2, 3$. Choose a complex embedding carrying the B_i above the linear residual root to B. Subtracting its term from each row leaves right sides of modulus at most $1 + 1/A$. Cramer's rule bounds each of the two other $|1/\gamma_i|$ by $1 + 1/A$. Thus $A < 3$, since each small $|1/\gamma_i|$ is at least $A^{17/40}$ and $3^{17/40} > 4/3$. Consequently $w = N(B) = 1$ since $0 < w \le B^{3/20} < 9^{3/20} < 2$. The small conjugates of B now have modulus $B^{-1/2}$, so $\sqrt{A} \le 1 + 1/A$, which implies $A < \sqrt{5}$ (at $A = \sqrt{5}$, $5^{1/4} > 1 + 1/\sqrt{5}$). Consequently $B < 5$. As $w = 1$, the reduction of F modulo three equals $(X - 1)(X^2 + 1)$ because the quadratic roots multiply to one and the product of all roots is one. Hence $u \equiv v \equiv 1 \pmod 3$. The elementary symmetric sums give $-1 < u < 7$ and $|v| < 2\sqrt{5} + 1 < 6$. Among $u \in \{1, 4\}$ and $v \in \{-5, -2, 1, 4\}$, the tests $F(1) < 0 < F(5)$ and $\operatorname{disc}(F) < 0$ leave precisely $(u, v) = (1, -2), (4, -2), (4, 1)$.

For these three cubics let $T_n = \sum_i B_i^n$. We eliminate each at modulus 27. In $(\mathbb{Z}/27)[X]/F$, the class x is a unit. Modulo three F is squarefree and its roots belong to $\mathbb{F}_9^\times$, so $x^8 = 1$ modulo three, and two cubings suffice to lift this to $x^{72} = 1$ modulo 27. In particular the claimed exponent 216 is valid. Thus the traces T_{m_k} modulo 27 for all sufficiently large k depend only on r mod 16 and k mod 4, since $216 = 8 \cdot 27$ and one may compute m_k using $r3^k - 1$ mod 432. The following standard-library calculation checks that some recurrent phase is neither sign in each case.

```
for u,v in ((1,-2),(4,-2),(4,1)):
    T=[3,u,(u*u-2*v)%27]
    while len(T)<216:
        T.append((u*T[-1]-v*T[-2]+T[-3])%27)
    for r0 in range(1,16,2):
        residues=[T[((r0*3**k-1)//2)%216] for k in range(6,10)]
        assert any(t not in (1,26) for t in residues)
```

This contradicts the congruence for P_k, proving that F is irreducible modulo three.

5 The two cases at thirteen

Case $13 \nmid r$. Write μ_n for the group of nth roots of unity. The three residual B_i form one Frobenius orbit in $\mathbb{F}_{27}^\times$. If $13 \nmid r$, this leads to a contradiction. Write the Teichmüller lifts for γ_i as $\omega_i = s_i t_i$, with $s_i \in \mu_4$ and $t_i \in \mu_{13}$. The squares s_i^2 are equal, since they are the sign components of the Frobenius-conjugate lifts of B_i. Changing the signs of γ_i when necessary makes all s_i equal to a common s. This is legitimate because every $r3^k - 1$ is even. The 13-components t_i^2 form a Frobenius orbit, and squaring is a bijection on μ_{13}. Thus, after indexing, the t_i are t, t^3, t^9, where t is primitive because the residual B_i have degree three.

Divide equation (2) at $j = 0, 1, 2$ by s^{r3^j}. The coefficient rows for the three unknowns $1/\gamma_i$ are the three cyclic shifts of (η, η^3, η^9), with $\eta = t^r$ primitive. To check that all its Fourier factors

are nonzero, take $w \in \mu_3$. The expression $\eta + w\eta^3 + w^2\eta^9$ is the evaluation at η of a nonzero polynomial of degree nine over $\mathbb{Q}(\mu_3)$, whereas the relative degree of η is $\varphi(39)/\varphi(3) = 12$. The right sides and all coefficients belong to $\mathbb{Q}(\mu_{52})$, so the solution does too. In particular the real cubic field $\mathbb{Q}(B)$ lies in an abelian extension of $\mathbb{Q}$ and is Galois. As a real Galois field it is totally real. This contradicts the nonreal conjugate pair established above. Consequently the minimum is transcendental for every odd r with $13 \nmid r$.

Case $13 \mid r$**.** The residual roots of B are now a Frobenius orbit in $\mathbb{F}_{27}^{\times}$, of order dividing 26. Because $13 \mid r$, the three v_i^2 are equal to a common sign s (all are signs, and Frobenius permutes them). Put $y_i = v_i/\gamma_i$. By (2), $\sum y_i = \pm 1$ and $y_i^2 = s/B_i$. Any Galois automorphism of a normal field containing these numbers sends each y_i to some $\pm y_{\pi(i)}$. A sign change would give a nonempty vanishing subsum of $\sum y_i$, by subtracting the two conjugate equalities. Neither one nor three terms can vanish, and two cannot vanish since the B_i are distinct. Thus there are no sign changes. In particular $y_1 \in \mathbb{Q}(B)$, which is real at the embedding with $B > 1$, giving $s = 1$. Consequently $A = \pm y_1^{-1}$ is cubic Pisot and $\mathrm{Tr}(A^{-1}) = \pm 1$. Write its minimal polynomial as $X^3 - aX^2 + bX - c$. The trace of the inverse is b/c, so $b = \pm c$. The conjugate bound gives $|N(A)| \le A^{3/20}$ and its nonreal conjugate pair gives positive norm. Finally the residues of the three B_i are distinct of degree three, so the residue of A also has degree three (as $A^2 = B$). Its own minimal polynomial is therefore irreducible over $\mathbb{F}_3$. Traces of the powers along the unchanged clock agree eventually with the prime floors because $A^2 = B$ and both generate the same cubic field.

6 A coefficient bound from an early prime floor

Suppose now that $13 \mid r$. Write $F(X) = X^3 - aX^2 + bX - c$ for the polynomial of the minimum A. Write T_m for its trace at exponent m. Every trace T_{r3^k-1}, including the early ones, is odd. Indeed, eventually they are odd primes, and $b = \pm c$. Modulo two, if c is even the trace at $m \ge 1$ is a^m. If a, c are both odd, $F = (X+1)^3$ in characteristic two and all these traces equal one. If a is even and c odd, $F = X^3 + X + 1$ modulo two and the roots have order seven. Then T_{r3^k-1} modulo two is periodic from the first index because 3 is a unit modulo seven. The same observation shows that a is odd when $7 \nmid r$. Indeed, if a is even we must have c odd. Since 3 generates $(\mathbb{Z}/7)^{\times}$, there is a phase with $r3^k - 1 \equiv 1 \pmod 7$, giving the even parity of a and a contradiction.

Set $N = 3r - 1$ if $r \equiv 3 \pmod 4$, and $N = 9r - 1$ otherwise, so $4 \mid N$. Write the small conjugates of A as $\sqrt{q}e^{\pm i\theta}$, with $q = c/A \le A^{-17/20}$ and $0 < \theta < \pi$. The trace of $1/A$ is ± 1, whence $\cos\theta = \sqrt{q}(\pm 1 - 1/A)/2$. If $A \ge (N/2)^{40/17}$, then $A > 4$ and $t = |\cos\theta| < 5/(4N) < 1/2$. Thus $N|\theta - \pi/2| = N \arcsin(t) < 6Nt/5 < 3/2$. As N is divisible by four, $\cos(N\theta) > 0$, and

$$0 < T_N - A^N = 2q^{N/2}\cos(N\theta) \le 2(2/N)^N < 1.$$

Therefore $\lfloor A^N \rfloor = T_N - 1$ is even and larger than two, contradicting its required primality at the original index. We conclude

$$A < (N/2)^{40/17}, \qquad c < (N/2)^{6/17}.$$

This argument uses no floor-to-trace identification at early indices.

The sum of the three conjugates gives $A - 2 < a < A + 2$. Since $A > 1$, the possible integer values lie between zero and $\lfloor (N/2)^{40/17} \rfloor + 2$, inclusive. We may also include $c = \lfloor (N/2)^{6/17} \rfloor$ when forming an outer bound. The conditions $3 \nmid c$, $b = \pm c$, irreducibility modulo three, and the parity restrictions above define the finite grid used below.

7 A finite covering of cubic coefficients

Let $f = X^3 - aX^2 + bX - c$ be a cubic in the grid. For a prime $p \geq 5$ with $p \nmid c$, the class x of X is invertible in $R = \mathbb{F}_p[X]/(f)$ and satisfies $x^{E_p} = 1$, where

$$E_p = p(p^2 - 1)(p^2 + p + 1).$$

In the reduced quotient each residue-field multiplicative group has order $p^j - 1$ for $1 \leq j \leq 3$, all dividing E_p/p. The nilradical J satisfies $J^3 = 0$, so $u^p = 1$ for every $u \in 1 + J$. Let $e = v_3(E_p)$. If $k \geq e$ and $\mathrm{Tr}_R(x^{r3^k - 1}) = 0$, take any $h > 0$ with $3^h \equiv 1 \pmod{E_p/3^e}$. The difference of exponents at $k + jh$ and k is $r3^k(3^{jh} - 1)$, divisible by E_p. Therefore $p \mid T_{r3^{k+jh}-1}$ for all $j \geq 0$. Since A is Pisot, the traces grow to infinity. A single such certificate excludes the entire coefficient residue class $a \pmod p$ for these b, c.

The checker in Appendix A searches $p \leq 1000$ and $\max(1, e) \leq k \leq 100$. It performs polynomial multiplication in the basis $1, x, x^2$, computes the algebra trace from $\mathrm{Tr}_R(1) = 3$, $\mathrm{Tr}_R(x) = a$, and $\mathrm{Tr}_R(x^2) = a^2 - 2b$, and uses $x^{r3^{k+1}-1} = x^2(x^{r3^k-1})^3$. Primes dividing c are always skipped. For each fixed pair (b, c) the residue classes obtained in this way are removed from the grid. The surviving count is an exact integer calculation. An independent check used powers of the 3×3 companion matrix modulo each witness prime and separately covered the coefficient intervals. Both calculations gave the same counts for all 38 multipliers.

8 What one residue class certifies

For $r = 273$ the first exponent divisible by four is $N = 9r - 1 = 2456$. The grid has $a \leq 18\,565\,304$, $c \leq 12$, and $3 \nmid c$. As $7 \mid 273$, even a must be retained when c is odd. Consider, for example, $f = X^3 - X^2 - X - 1$. At $p = 5$ and $k = 6$ its exponent is $273 \cdot 3^6 - 1 = 199\,016$. Reduction by $x^3 = x^2 + x + 1$ gives

$$x^{199\,016} = 2 + 3x + 2x^2 \quad \text{in } \mathbb{F}_5[X]/(f).$$

Its trace is $2 \cdot 3 + 3 \cdot 1 + 2 \cdot 3 = 15 \equiv 0 \pmod 5$. Here $E_5 = 3720$, $v_3(E_5) = 1$, and $3^{60} \equiv 1 \pmod{1240}$. The same prime divides every trace at $k = 6 + 60j$, $j \geq 0$. This rules out the whole class $a \equiv 1 \pmod 5$ for $b = -1$, $c = 1$. The computation in the appendix performs the same operation with the witness prime and index chosen separately for each class.

Multiplier(s)	Triples in the grid	Rules used	Largest prime
13	21 153	388	53
117	7 576 020	2 562	103
273	111 391 828	4 188	179
949	2 265 148 054	11 577	173
All 38	13 197 086 314	181 239	241

Counts are per multiplier and then summed. In every row the number of uncovered triples is zero. Only eventual equality of floor and trace enters the exclusion, even when the first residue witness occurs early. Together with the case $13 \nmid r$ this proves Theorem 1.

The size limit 1000 specifies the range of the exhaustive verification. For a further odd multiple of thirteen the reduction and the height bound still apply, but the resulting finite grid needs its own covering. Even multipliers, including $r = 2$, require a different reduction at the outset because C_k is odd. The classical Mills minimum has the different exponent sequence $(3^k)_{k \geq 1}$.

References

[1] K. Saito, *Transcendency of variants of Mills' constant*, Ramanujan J. **70** (2026), article 69. doi:10.1007/s11139-026-01443-0. Theorem numbering in this paper refers to arXiv:2508.16068v3.

[2] Keio, *Ternary trace dynamics and the transcendence of shifted Mills constants*, Verifiable Mathematical Reasoning Corpus, manuscript 064, 25 October 2026, Theorem B and Section 7.2.

[3] Keio, *Shifted Mills Constants and Cubic Pisot Trace Dynamics*, Verifiable Mathematical Reasoning Corpus, manuscript 075, 27 October 2026, in particular the discussion of $r = 273$ following the proof for $r = 195$. Both corpus references use the 2 November 2026 repository revision.

A Reproducing the finite covering

The following C++17 program uses only the standard library. The five output columns are r, grid size, number of covering rules, largest witness prime, and number left. Pass `273` to the executable for the single case or pass no argument for all 38. For instance, the following commands extract the listing from this TeX source and execute it.

```
awk '/^%% BEGIN CHECKER$/ {f=1;next} /^%% END CHECKER$/ {f=0}
     f && $0 !~ /^\\(begin|end)\{lstlisting\}/ {print}' \
     Shifted_Mills_Minima.tex > check.cpp
c++ -O2 -std=c++17 check.cpp -o check
./check 273
./check
```

The covering loop keeps a list of progressions modulo a common width. At a new prime p, each progression splits into at most p classes. It retains only those whose trace has no witness at this prime. Once the width exceeds the coefficient bound, each class is a singleton. This explains why the memory cost is smaller than the initial grid size. The two integer root bounds are computed by comparing exact powers in radix 10^8. The anticipated final line is

```
total 13197086314 181239 241 0
```

```cpp
// C++17. With no argument, checks all 38 multipliers. With e.g. 273, checks one.
#include <algorithm>
#include <array>
#include <cstdint>
#include <cstdlib>
#include <iostream>
#include <vector>

using Int = std::int64_t;
using Poly = std::array<int,3>;

std::vector<int> exact_power(int base, int exponent) {
    // The digit count followed by the most significant digits compares lexically.
    constexpr int radix = 100000000;
    std::vector<int> digits{1};
    for (int i=0; i<exponent; ++i) {
        Int carry=0;
        for (int& d: digits) {
            carry += Int(d)*base;
            d = int(carry%radix);
```

```
            carry /= radix;
        }
        while (carry) {
            digits.push_back(int(carry%radix));
            carry /= radix;
        }
    }
    digits.push_back(int(digits.size()));
    std::reverse(digits.begin(),digits.end());
    return digits;
}

int bound(int base, int power) {
    auto target=exact_power(base,power);
    int lo=0, hi=1;
    while (exact_power(hi,17)<=target) hi*=2;
    while (hi-lo>1) {
        int mid=lo+(hi-lo)/2;
        if (exact_power(mid,17)<=target) lo=mid;
        else hi=mid;
    }
    return lo;
}

bool prime(int n) {
    for (int d=2; d*d<=n; ++d) if (n%d==0) return false;
    return n>=2;
}

struct Cubic {
    int a,b,c,p;
    int mod(Int n) const { return int((n%p+p)%p); }
    Poly multiply(Poly u, Poly v) const {
        // Reduce the degree-four product using X^3=aX^2-bX+c.
        Int h4=Int(u[2])*v[2];
        Int h3=mod(Int(u[1])*v[2]+Int(u[2])*v[1]+a*h4);
        return {mod(Int(u[0])*v[0]+c*h3),
                mod(Int(u[0])*v[1]+Int(u[1])*v[0]-b*h3+c*h4),
                mod(Int(u[0])*v[2]+Int(u[1])*v[1]+Int(u[2])*v[0]+a*h3-b*h4)};
    }
    int witness(int r, int depth) const {
        const Poly x2{0,0,1};
        Poly x{0,1,0}, u{1,0,0};
        for (int n=3*r-1; n; n/=2) {
            if (n%2) u=multiply(u,x);
            x=multiply(x,x);
        }
        for (int k=1; k<=100; ++k) {
            Int trace=3*u[0]+Int(a)*u[1]+Int(a*a-2*b)*u[2];
            if (k>=depth && mod(trace)==0) return k;
            u=multiply(x2,multiply(u,multiply(u,u)));
        }
        return 0;
    }
};

struct Counts { Int grid=0, rules=0, left=0; int largest=0; };

void check_row(int r,int b,int c,int A,Counts& counts) {
    // A representative j denotes all j+width*t in [0,A], t>=0.
    std::vector<int> live;
    Int width=6;
    for (int d=0; d<6; ++d) {
        if (d%2==0 && (r%7!=0 || c%2==0)) continue;
```

```
        bool root=false;
        for (int x=0; x<3; ++x)
            if ((x*x*x-d*x*x+b*x-c)%3==0) root=true;
        if (!root) { live.push_back(d); counts.grid+=(A-d)/6+1; }
    }
    for (int p=5; p<=1000 && !live.empty(); p+=2) {
        if (!prime(p) || c%p==0) continue;
        int depth=(p%3==1);
        for (int t=p*p-1; t%3==0; t/=3) ++depth;
        std::vector<int> cache(p,-1), next;
        for (int j:live) {
            Int last=std::min<Int>(A,j+(p-1)*width);
            for (Int a=j; a<=last; a+=width) {
                int v=int(a%p);
                if (cache[v]<0) {
                    cache[v]=Cubic{v,b,c,p}.witness(r,depth);
                    if (cache[v]) {
                        ++counts.rules;
                        counts.largest=std::max(counts.largest,p);
                    }
                }
                if (cache[v]==0) next.push_back(int(a));
            }
        }
        width=std::min<Int>(Int(A)+1,width*p);
        live.swap(next);
    }
    for (int j:live) counts.left+=(A-j)/width+1;
}

int main(int argc,char** argv) {
    int only=argc==2?std::atoi(argv[1]):0;
    if (argc>2 || (argc==2 && (only<13 || only>1000 || only%26!=13))) return 2;
    Counts all;
    for (int r=13; r<=1000; r+=26) {
        if (only && only!=r) continue;
        int N=(r%4==3?3:9)*r-1;
        int A=bound(N/2,40)+2, C=bound(N/2,6);
        Counts row;
        for (int c=1; c<=C; ++c) if (c%3)
            for (int b:{-c,c}) check_row(r,b,c,A,row);
        std::cout<<r<<" "<<row.grid<<" "<<row.rules<<" "
                 <<row.largest<<" "<<row.left<<std::endl;
        all.grid+=row.grid; all.rules+=row.rules; all.left+=row.left;
        all.largest=std::max(all.largest,row.largest);
    }
    std::cout<<"total "<<all.grid<<" "<<all.rules<<" "
             <<all.largest<<" "<<all.left<<std::endl;
    return all.left!=0;
}
```